

PORTERVILLE, South Africa

WMO: 687170

Lat: 33.013S

Lon: 18.977E

Elev: 122

StdP: 99.87

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.9	4.0	0.5	4.0	12.2	1.6	4.3	10.8	7.4	12.5	6.7	13.1	0.7	90	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.7	36.8	20.3	35.0	19.7	33.2	19.3	21.5	33.7	20.7	32.1	20.0	30.9	2.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.8	24.7	16.8	12.2	23.8	16.2	11.7	23.6	63.1	33.5	60.1	32.3	57.9	31.0	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.2	5.2	1.2	41.2	0.9	0.8	0.5	41.8	0.0	42.3	-0.5	42.8	-1.1	43.4		
			0.5	23.0	0.7	0.6	0.0	23.4	-0.5	23.8	-0.9	24.1	-1.4	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	25.1	25.0	23.0	19.5	15.7	12.8	11.9	11.9	14.1	17.9	20.5	23.4
	DBStd	5.69	3.15	3.26	3.21	3.46	2.79	2.42	2.54	2.15	2.66	3.43	3.32	3.10
	HDD10.0	18	0	0	0	0	0	2	8	6	1	0	0	0
	HDD18.3	886	1	1	2	26	93	167	199	200	131	49	17	2
	CDD10.0	3071	468	421	402	284	176	87	69	64	124	246	315	415
	CDD18.3	897	210	188	147	60	11	2	1	0	3	37	82	158
	CDH23.3	8551	2108	1832	1217	498	80	14	9	5	61	431	772	1525
	CDH26.7	3834	1053	905	514	173	13	2	1	0	18	158	306	692
Wind	WSAvg	1.7	2.0	1.8	1.6	1.5	1.7	1.9	1.7	1.6	1.4	1.6	1.8	2.0
Precipitation	PrecAvg	441	10	11	15	33	60	77	69	69	36	22	23	15
	PrecMax	644	49	70	62	67	139	207	175	136	95	60	57	51
	PrecMin	294	0	0	1	5	5	10	22	26	7	2	0	1
	PrecStd	95	11	18	15	17	37	40	41	27	25	14	16	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.1	38.5	37.2	34.3	28.5	25.5	24.6	23.8	28.4	35.1	35.8	37.7
		MCWB	21.1	21.0	20.3	18.6	16.4	15.4	16.1	16.3	17.3	19.2	19.2	20.4
	2%	DB	36.8	36.1	33.9	31.0	25.6	21.7	21.1	20.6	24.7	30.9	32.7	34.9
		MCWB	20.6	20.2	19.4	18.2	15.5	13.9	14.6	14.7	15.8	17.8	18.2	19.3
	5%	DB	34.5	34.1	31.7	28.5	23.3	19.1	18.6	18.2	22.1	28.0	30.3	32.8
		MCWB	20.0	19.8	18.7	17.3	15.0	13.4	13.3	13.4	14.7	16.6	17.5	18.9
	10%	DB	32.2	32.0	29.6	26.2	20.8	17.0	16.6	16.3	19.6	25.4	28.0	30.7
		MCWB	19.3	19.4	18.3	16.8	14.4	12.7	12.3	12.2	13.8	15.7	16.8	18.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.6	22.7	21.5	20.0	17.8	16.1	17.0	17.2	18.4	20.0	20.4	21.4
		MCDB	36.0	34.9	33.6	31.3	24.8	22.2	22.9	22.6	26.7	32.6	31.9	35.0
	2%	WB	21.5	21.6	20.3	19.0	16.5	14.7	15.1	15.2	16.5	18.5	19.1	20.2
		MCDB	34.4	32.8	31.5	28.2	23.2	19.4	20.6	19.9	23.6	28.9	29.9	32.3
	5%	WB	20.6	20.8	19.5	18.2	15.6	13.9	13.7	13.6	15.2	17.3	18.2	19.5
		MCDB	32.1	31.3	29.6	26.5	21.1	17.5	17.6	17.5	21.3	26.1	28.3	30.6
	10%	WB	19.8	20.0	18.7	17.2	14.8	13.2	12.7	12.5	14.1	16.2	17.4	18.7
		MCDB	30.3	30.0	27.8	24.9	19.7	16.1	15.8	15.7	19.0	23.7	26.2	28.7
Mean Daily Temperature Range	5% DB	MDBR	13.7	13.4	12.9	12.2	9.9	9.4	10.1	9.8	11.4	13.7	13.7	13.7
		MCDBR	17.0	16.3	15.8	16.0	14.6	13.6	14.1	14.2	16.4	18.5	17.9	17.0
		MCWBR	4.9	4.6	4.9	5.6	6.6	7.7	9.0	9.0	8.3	7.3	6.0	5.3
	5% WB	MCDBR	15.4	13.9	14.3	13.7	12.3	10.8	12.5	13.1	15.0	17.1	16.1	15.4
	MCWBR	4.8	4.3	4.8	5.3	6.0	6.7	8.3	8.5	8.1	7.0	5.7	5.0	
Clear-Sky Solar Irradiance	taub	0.336	0.337	0.333	0.318	0.310	0.303	0.306	0.330	0.341	0.352	0.332	0.336	
	taud	2.481	2.483	2.485	2.534	2.548	2.542	2.523	2.449	2.387	2.373	2.471	2.477	
	Ebn at Noon	1002	981	948	910	864	845	859	881	924	953	999	1005	
	Edh at Noon	116	112	104	89	78	73	78	95	113	124	117	118	
All-Sky Solar Radiation	RadAvg	8.41	7.58	6.17	4.53	3.21	2.71	3.02	3.76	5.17	6.70	7.90	8.49	
	RadStd	0.22	0.20	0.19	0.23	0.21	0.17	0.17	0.18	0.22	0.32	0.23	0.28	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	+0.35	N/A	N/A	+0.21	N/A	-2	-74	+111	+61

Nomenclature: See separate page